Environmental and Health Economics:

Do Cigarette Taxes Affect Birthweight?

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Abstract

Maternal smoking is one of the major risk factors for low birth weight in the United States, linked to the long-term development of newborns. Thus, cigarette taxation is publicly used to restrain smoking behavior among pregnant women. This research study aims to examine the impact of cigarette tax policies and maternal smoking on infant birth weight. Using the US national birth dataset from 1988, we estimate a linear regression model in which birth weight is regressed on maternal daily cigarette consumption and cigarette tax in dollars, while holding other factors such as family income, parental education, race, baby gender, etc., as the unobserved variables. We find that each additional dollar in cigarette tax will lead to a significant increase in birth weight. Additionally, an additional cigarette consumed by pregnant women is associated with a reduction in infant birth weight. Our result shows the effectiveness of cigarette taxation on birth weight and emphasizes the importance of lower cigarette consumption among pregnant women.

Introduction

One of the leading factors of death, disease, and disability is cigarette smoking, posing a major threat to public health and well-being. The use of tobacco peaked in the 90s and has remained widespread among Americans ever since, including pregnant women. Concurrently, smoking habits during pregnancy have been reported to be associated with the risks of birth complications, infant death syndrome, low birth weight, etc. For this reason, reducing the number of women who smoke during pregnancy has become an urgent public goal. Accordingly, the government imposed a high cigarette tax, which was one of the solutions to decrease the smoking rates among pregnant women. Numerous research studies have sought to estimate the overall impact of cigarette tax policy on birth outcomes using repeated observations and data. In this research project, we want to conduct an empirical analysis to examine the relationship between the predictor variables (cigarette taxes and number of cigarettes pregnant women used per day) and the response variable (birth weight by ounce) in the United States during 1988-2021.

Literature Review

According to the research journal "Can higher cigarette taxes improve birth outcomes?" by William Evans and Jeanne Ringel (1999), they estimated that approximately 20% of all low birth weight was caused by smoking. Furthermore, despite the slow fetal growth can be caused by various biological factors, they discovered that cigarette taxes do effectively alter the smoking behavior of pregnant women. William and Jeanne both stated that the relationship between cigarette taxes and birth outcomes is positively correlated. Their strategy was to divide the research into two distinct approaches: the impact of taxes on maternal smoking and the impact of taxes on smoking behavior and birth weight. For the first analysis, instead of focusing on the

national level, they researched the state tax rate and levels of cigarette consumption in 1988. They found that there was no statistically significant relationship between cigarette tax changes and the growth rate of smoking in the state. For the second analysis, they found a significant percentage of women who smoke before pregnancy quit once they learn they are pregnant. Therefore, they narrowed down to apply the relevant tax rate during the month of conception and women in 15-44 years of age. The outcome of interest in this study was infant birth weight in grams; while smoking behavior (number of cigarettes smoked per day) and monthly state tax were independent factors. They included additional indicators such as mother's age and race, prenatal care, and marital status.

As a result, they assessed that "a one-cent real tax increase would lead to approximately a 0.08 percentage point reduction in the mean rate of smoking participation" (Evans and Ringel, 1999, p.11). The estimates indicate that the effect of the cigarette tax on smoking participation is robust. Moreover, as the researchers expected, they found that "a one-cent increase in the state tax rate on cigarettes increases average birth weight by 0.16g" (Evans and Ringel, 1999, p.14-15). They also provided sample data, showing that an increase of $1.10 tax would reduce maternal smoking by 32% and reduce low birth rates by 5%. Since pregnant women are responsive to taxes changes, the model demonstrated that increased in cigarette tax rates have a beneficial impact on mean birth weight.

In addition, the researchers found a strong association between higher cigarette taxes and decreases in infant mortality. From 1999 to 2010, the mean state infant mortality rate decreased from 7.3 to 6.2 per 1000 births. At the same time, the state means inflation-adjusted cigarette tax increased from $0.84 to $2.37 (Patrick, Warner, Pordes, and Davis, 2016). They concluded that every $1 increase in cigarette tax per pack was associated with a change in infant mortality rate of -0.032 in the population overall (750 fewer infant deaths per year). Overall, both studies

demonstrated that cigarette tax policies are effective in easing public concerns regarding the risks of infant mortality and low birth weight.

Econometric Model

On the research mentioned above we find that there are many factors impacting child birthweight other than cigarette taxation, such as parental education, family income, and even birth parity. The given research journal "Can cigarette taxes improve birth outcomes?" does not act on these parameters for two reasons: not having enough collected data and being able to account for these values through the error term *e* (Evans and Ringel, 1999, p.8-9). Although this information may be relevant to child birthweight, including the parameters in the regression model could be redundant for the context of our research. Like cigarette taxation, parameters such as family income and parental education affect our value of interest indirectly. Such variation is, in effect, attributed to the error term.

Initiating our research, we built a multiple linear regression model with relevant parameters to our case. The objective of our study is to investigate how cigarette taxes impact birthweight, therefore defining birthweight (*bwght*) as our response variable Y. Our first coefficient, β0, accounts for the expected low birthweight outcome when there is no cigarette consumption during pregnancy and no cigarette tax is imposed. Our second coefficient, β1, models cigarette taxes within the home state where pregnancy occurs (*cigtax*). Our data considers the location of the pregnancy and matches that to smoking mothers within their respective area, in this case their state of residence. Our third and final parameter, β2, is defined by the number of cigarettes smoked per day during a pregnancy (*cigs*). It is fundamental to add cigarette consumption during pregnancy due to its direct correlation with cigarette tax. Cigarette taxes affect the consumption of these, following to affect infant birthweight.

$$bwght = \beta_0 + \beta_1 cigtax_1 + \beta_2 cigs_2 + e$$

Using Stata, we run our multiple linear regression model to find our estimated coefficients. Our first coefficient, β0, is given a value of 117.11 units. With this coefficient, we assume that when the mother does not smoke cigarettes during pregnancy, and cigarette taxes are not imposed, the estimated birthweight of the infant will be 117.11 ounces, which is considered normal in a newborn baby. We find that when cigarette consumption remains constant, a one dollar increase in cigarette tax will lead to a 0.14 ounce increase in infant birthweight. Vice versa, when cigarette taxes remain constant, an additional cigarette smoked during pregnancy will lead to a 0.52 ounce decrease in infant birthweight. As we mentioned earlier, our error term accounts for the variation caused by other indirect factors that may affect birthweight.

$$\widehat{bwght} = 117.11 + 0.14cigtax - 0.52cigs + e$$

Data Description

The summary statistic, as seen on *Figure ii*, shows the average infant weight is about 119 ounces, though the data reveal substantial variability, with some cases reflecting significantly low birth weights based on 1,388 observations. The first column shows significant variability in infant health outcomes, underscoring the importance of contributing risk factors like maternal smoking. Birth weight ranges from as low as 23 ounces to a maximum of 271 ounces. The minimum and maximum birth weights, measured in ounces, reflect instances of severely low birth weight and exceptionally high birth weight. The standard deviation of 20.35 ounces suggests a moderate level of variation in birth weights.

On average, pregnant women smoked approximately 2 cigarettes per day, but the maximum number of cigarettes pregnant women smoked was up to 50 cigarettes. A broad segment of women did not smoke at all during pregnancy (min = 0). This statistical report

indicates a high standard deviation relative to the mean, suggesting that smoking habits vary widely among pregnant women–some smoked a lot more than usual, while many abstained from smoking entirely.

The mean state cigarette tax among the observed sample is $19.55, varying by state, suggesting a moderate level of taxation overall. Specifically, tax rates range from a minimum of $2 to a maximum of $38, highlighting a disparity in how different states regulate tobacco through policy. Such a considerable range and variation of cigarette taxes are useful for examining how the differences in taxation may impact maternal smoking and, in turn, affect birth weights.

Empirical Results

Using our given program, Stata, we run our multiple linear regression mode to determine levels of significance and other results relevant to our research question. As seen in *Figure i*, the given regression model reports crucial values in our research questionnaire such as t-statistics and p-values, standard errors, r-square, among others. To find the significance levels of each coefficient, we follow t-statistic and p-value guidelines. For a coefficient to be considered statistically significant, its t-statistic must exceed 1.96 in absolute value, and be should not exceed 0.05 in its corresponding p-value. As *Figure i* illustrates, the t-statistic of our second coefficient, β1 or *cigtax*, is 1.97, which indicates that cigarette taxes are significantly related to our response variable. Furthermore, the p-value of cigarette taxes is 0.049, also proving significance to infant birthweight. We also see that cigarette consumption, *cigs*, has a t-statistic value of -5.74, being greater than 1.96 in absolute value. Cigarette consumption also carries a p-value of 0.00, showing significance at the 5% level to the response variable. Both of our coefficients are significant to infant birthweight, allowing us to reject the null hypothesis.

By running our regression model in Stata, we were able to determine values such as the r-squared of the model. We find that the r-squared of this regression model is 0.0255, also seen as 2.55%. Using the r-squared, we know that 2.55% of the variation in birthweight can be explained by cigarette tax and cigarette consumption during pregnancy. As we know, cigarette consumption and taxes are not the sole factors impacting birthweight, being proven by its low R-squared. As a matter of fact, 97.45% of the variation is explained by other factors, those of which our error term accounts for.

Limitations

Although our model includes statistically significant variables, it encounters several constraints, which cause our findings to be less precise. First, women may underreport their cigarette consumption in surveys, making our β2 less negative than it truly is. It means that the negative influence of smoking on infant weight is underestimated. Second, our model did not account for other factors that affect infant birthweight, such as parental education, family income, gender, prenatal care, etc. The exclusions of these variables lead to omitted variable bias, potentially misinterpreting the estimated effect of included predictors. Third, our model has a low R-squared (2.5%), our model explains only a small portion of variation in the dependent variable (*bwght)*. Although our model shows that maternal smoking and cigarette tax are statistically significant, both have a small effect compared to other influential variables I previously mentioned. If possible, in the future direction of our research, we should include additional control variables to enhance the percentage of variation in birthweight. Last, the results of our analysis indicate an association between cigarette taxation and birthweight; however, the magnitude of this effect may not remain stable in the long term. An increase in cigarette taxes may slightly or heavily impact the outcome.

Conclusion

Based on the supporting evidence from the p-value and t-statistics, we reject the null hypothesis, which indicates that both cigarette taxation and maternal smoking have a statistically significant effect on birthweight. Furthermore, we found a strong association and positive correlation between cigarette taxes and infant birthweight. Our findings are consistent with previous research studies, suggesting that each additional dollar in cigarette tax will lead to a significant increase in birthweight and a decrease in maternal smoking. This implies that cigarette tax policies not only act as a restraining factor to maternal smoking but also serve as an effective public health tool in improving birth outcomes.

References

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Appendix A

Figure i:

Source	SS	df	MS			
				Number of obs	=	1,388
				F(2, 1385)	=	18.09
Model	14629.1205	2	7314.56027	Prob > F	=	0.0000
Residual	559982.599	1,385	404.319566	R-squared	=	0.0255
				Adj R-squared	=	0.0241
Total	574611.72	1,387	414.283864	Root MSE	=	20.108

bwght	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
cigs	-.5190009	.090436	-5.74	0.000	-.6964073	-.3415944
cigtax	.1364802	.0692886	1.97	0.049	.0005582	.2724022
_cons	117.1142	1.465397	79.92	0.000	114.2396	119.9889

Figure ii:

Variable	Obs	Mean	Std. Dev.	Min	Max
bwght	1,388	118.6996	20.35396	23	271
cigs	1,388	2.087176	5.972688	0	50
cigtax	1,388	19.55295	7.795598	2	38